

Naveen Prashanna Gurumurthy

+1 (945) 527-5193 | gnaveen1509@gmail.com | [LinkedIn](#) | [Portfolio](#)

Summary

AI/Machine Learning Engineer with 3+ years of experience building production-grade AI/ML systems across quantitative finance, generative AI & browser automation. Expertise in Alpha Research, LLM engineering, RAG, multi-agent systems, fine-tuning, reinforcement learning, MLOps & data pipelines, with hands-on experience across Azure, LangGraph, CrewAI, MLflow, PyTorch & Scikit-learn. Proven ability to translate applied research into scalable, reliable AI solutions, combining strong Python, cloud, backend, data engineering & quantitative modeling skills.

Technical Skills

- **Languages:** Python, C++, JavaScript, TypeScript, React.js, SQL, R, MATLAB, Bash
- **AI / ML & LLM Engineering:** PyTorch, TensorFlow, Scikit-learn, Mistral, LangChain, LangGraph, CrewAI, LLMs, RAG, QLoRA, Fine-Tuning, Prompt Engineering, Vector Search, Graph RAG, Embeddings, Multi-Agent Systems, Pydantic, Reinforcement Learning
- **Data Engineering & Backend:** FastAPI, Polars, Pandas, NumPy, Delta Lake, Azure Blob, Neo4j, PostgreSQL, MongoDB, REST APIs
- **Cloud, MLOps & Tools:** AWS, Azure AI Foundry, Azure Automation, Docker, Jenkins, Databricks, MLflow, OpenAI Gym, Airflow

Professional Experience

Quantitative Researcher, Pharvision Advisers | USA

Jan 2026 – Present

- Engineered multi-factor machine learning models processing over **15 million** daily tick-level data points across global equities and macro datasets, utilizing point-in-time alignment to achieve Sharpe > 1.2 at low portfolio turnover.
- Orchestrated LLM-based multi-agent workflows using Langchain, MLflow, and W&B handling over **10,000** automated backtesting runs weekly across 50+ concurrent experiments, reducing model validation cycles from days to hours.
- Developed an agentic production RAG system serving live financial research, combining Neo4j GraphRAG (multi-hop traversal) and dense vector search (FAISS) with cross-encoder reranking to process complex queries across **500GB+** of documentation.
- Integrated a RAG evaluation framework utilizing LLM as a judge to benchmark response faithfulness, context relevance & retrieval accuracy.
- Established scalable data pipelines with Polars, Airflow, and Delta Lake ingesting **10+** vendor feeds concurrently (handling **50M+** rows daily) to expand quantitative research coverage without latency degradation.

AI Engineer, Kahana Group Inc | USA

Jul 2025 – Jan 2026

- Programmed production-ready LLM execution flows using Azure AI Foundry and Azure AI Services for an AI-native browser, orchestrating robust state transitions and prompt workflows to elevate reliability across complex multi-step user tasks.
- Architected an Azure-backed persistence layer using Azure Blob Storage scaling to manage browser state, sessions, bookmarks, and tabs with **sub-50ms** sync overhead under heavy state transitions.
- Instrumented real-time AI observability telemetry with Azure Monitor Application Insights, tracking token metering and usage analytics across architecture stress tests simulating **5M+** monthly LLM throughput and trim cloud cost overhead.
- Established automated evaluation harnesses using the Azure AI Evaluation SDK to analyze user interactions, model responses, and failure patterns at scale, boosting prompt optimization effectiveness and completion metrics.

Software Engineer (ML Systems), Quantitative Brokers | India

Jul 2022 - Feb 2023

- Built low-latency C++/Python inference runtimes and event-driven messaging pipelines executing predictive models with sub-millisecond dispatch ($< 800\mu s$) over high-throughput FIX protocol sessions processing millions of daily messages.
- Integrated real-time feature streaming and state tracking for multi-leg execution strategies, dynamically processing live Level-2 order book feeds across **40+** asset pairs to minimize market execution slippage under peak institutional loads.
- Engineered high-throughput PostgreSQL and Kafka telemetry pipelines capturing over **10GB** of streaming fills and model predictions daily, eliminating data leakage and accelerating offline feature engineering by **3.5x**.
- Extended the core FIX messaging platform and backend services with multi-leg trade support, optimizing low-level data-access layers to handle heavy-load financial traffic spikes seamlessly.

Data Scientist Intern, Big Data Science Research | Bangalore, India

Apr 2021 - Jun 2022

- Developed and deployed a machine learning pipeline using Python and scikit-learn to predict urban traffic patterns, enhancing model accuracy by 15% through the integration of geospatial data features.
- Implemented a novel map-matching algorithm to refine GPS data, significantly improving the precision of traffic flow models by 20%.

Projects

Generative AI Engineer, Personal AI Assistant Chatbot

Apr 2025 - May 2025

- Fine-tuned a Mistral LLM using QLoRA & deployed on cloud GPUs via vLLM, cutting memory by **40%** while retaining baseline performance.
- Integrated a high-throughput RAG pipeline with document embeddings, vector search, and semantic retrieval, optimizing context windows to deliver sub-second responses and cutting hallucinations by **78%**.

Deep RL Engineer, Trajectory-Aware Human Feedback for Hierarchical RL

Sep 2024 - Dec 2024

- Formulated a Hierarchical Reinforcement Learning (HRL) framework using trajectory-aware human feedback, subgoal generation, and policy optimization for complex long-horizon control environments across **500k+** simulation steps.
- Evaluated the framework in OpenAI Gym FetchReach, increasing task success by **10%** and reducing unnecessary exploration by $\sim 18\%$.

Quant Developer, Algorithmic Trading Engine

Jun 2024 - Aug 2024

- Engineered a Pandas-vectorized backtesting engine processing 10 years of high-frequency OHLC market data across **25+ million** rows, modeling transaction costs and slippage to stress-test institutional portfolio strategies.
- Implemented RSI mean-reversion models with walk-forward cross-validation and grid-search optimization, achieving an out-of-sample Sharpe ratio of ~ 0.9 with a **33%** reduction in maximum drawdown versus buy-and-hold benchmarks.

Education

Master of Science, Computer Science - Intelligent Systems | University of Texas at Dallas (UTD)

Aug 2023 - May 2025

B.Tech + M.Tech, Mechanical Engineering - Minor: AI & ML | Indian Institute of Technology (IIT) Madras

Aug 2017 - May 2022